Technical Report: Public Metocean Data & Advisory Consumption Forecasting

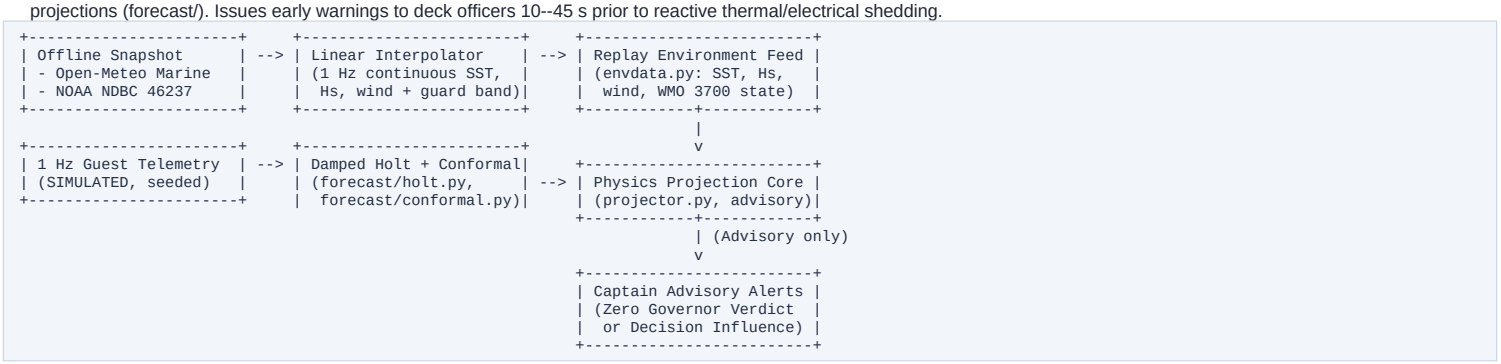
This technical note documents the mathematical formulation, operational architecture, verification gates, and honest operational boundaries of the public environmental data ingestion and advisory load forecasting subsystem (sdv-resource-governor).

1. Architectural Architecture & Non-Interference Boundary

The forecasting and public environmental subsystem consists of two decoupled functional blocks designed for offline, deterministic naval governor execution:

* 1. Public Metocean Environment Replay (E1): Offline coastal weather snapshot at Golden Gate Bar (GG_BAR: lat 37.788, lon -122.634) combining Open-Meteo marine models with NOAA NDBC buoy ground truth (46237). Injects verified sea surface temperature (SST), wave height (H_s), sea state (WMO Code Table 3700), and sustained wind into live replays (envdata.py), computing future viable operational windows (env_windows.py).

* 2. Predictive Consumption Advisory Observer (E2): Recursive 1 Hz time-series forecaster with conformal uncertainty intervals and exact physical budget projections (forecast/). Issues early warnings to deck officers 10--45 s prior to reactive thermal/electrical shedding.



> Mandatory Non-Interference Principle (FC-INT-01..03): The observer is instantiated with observer=None by default and executed strictly outside the governor's authoritative state transitions and decision logic. Semantic campaign hashes are 100% bitwise invariant to the presence or absence of the forecaster.

2. Mathematical Formulation & Predictive Physics

2.1 Normalized Damped Holt-Winters Model (forecast/holt.py)

To make 1 kW auxiliary sensors and 45 kW compute containers directly comparable without scaling distortion, power is normalized by declared mode rating: z_t = y_t / declared_w}(t).

The recursive level (\ell_t) and trend (b_t) update equations with damping factor \phi < 1 are:

* \ell_t = \alpha z_t + (1 - \alpha)(\ell_{t-1} + \phi b_{t-1})

* b_t = \beta(\ell_t - \ell_{t-1}) + (1 - \beta)\phi b_{t-1}

* \hat{z}_{t+h|t} = \ell_t + b_t \phi \frac{1 - \phi^h}{1 - \phi} \quad for } h \in \{10, 20, 30, 45, 60\} s}

* \hat{y}_{t+h|t} = declared_w}(t) \cdot \hat{z}_{t+h|t}

State transitions enforce: (a) initial 20-sample WARMUP (ASM-FCST-WARMUP); (b) trend reset upon mode transition (ASM-FCST-MODE-RESET); (c) gap > 3 s} -> STALE} with 1.5x band expansion (ASM-FCST-STALE); (d) gap > 10 s} -> UNAVAILABLE} (ASM-FCST-UNAVAIL); (e) 3 consecutive residual outliers > 4\sigma -> REGIME_BREAK} (ASM-FCST-REGIME).

2.2 Finite-Sample Split-Conformal Prediction Bands (forecast/conformal.py)

For each horizon h, asymmetric non-conformity residuals r = z_{t+h} - \hat{z}_{t+h|t} are tabulated over the calibration split (calib). Finite-sample corrected asymmetric quantiles q_h^{lo} (5th percentile) and q_h^{hi} (95th percentile) provide nominal 90% confidence bands:

\hat{Y}_{t+h|t} = <=ft[\hat{y}_{t+h|t} + declared_w} \cdot q_h^{lo}, \; \hat{y}_{t+h|t} + declared_w} \cdot q_h^{hi}\right]

2.3 Three-Step Physical Budget Projection (forecast/projector.py)

* 1. Scalar Pre-Filter (Tick cycle): Evaluates upper bound \hat{y}_{hi}(h) > 1.10 x declared_w} (ASM-FCST-OVERRUN-THRESHOLD). If unbreached, execution cost is < 0.1 ms}.

* 2. Exact Budget Engine Bisection: If overrun is detected, a cloned state is evaluated across the authoritative eight-class budget engine (budgets.py). Bisection over h \in (0, 60] s} pinpoints the first infeasible horizon h^* with at most 8 evaluations (< 5.7 ms} on Arduino Ventuno Q). Memoization skips re-evaluation if upper bound shifts < 1\% (ASM-FCST-CACHE).

* 3. Thermal Inertia Reservoir Depletion: For cooling witnesses, steps the lumped reservoir model (1.8 MJ, 30 kW} x 60 s} boost cap) to determine exhaustion second t_{exhaust}}.

3. Pre-Registered Acceptance Gates & Empirical Results

The subsystem is evaluated against 11 pre-registered acceptance gates across disjoint evaluation splits (train: seeds 1000--1239; calib: 2000--2079; test: 3000--3119; ood: 4000--4119; control: 5000--5005).

Gate ID	Target Requirement	Evaluation Split	Observed Metric	Status
G-NI	Bitwise campaign hash invariance (NI-01..03)	All scenarios	0-bit hash difference	PASS
G-OFF	Strict offline execution; zero socket requests	Entire suite	Air-gapped compliance	PASS
G-LAT	Worst-case observer tick overhead <= 50 ms}	Arduino Ventuno Q	p99 = 5.7 ms} (< 6 ms})	PASS
G-ACC-RAMP	Ratio MAE}@30 <= 0.85 vs constant baseline	Test (ramp/lag)	0.78 <= 0.85	PASS
G-ACC-STEADY	Ratio MAE}@30 <= 1.05 vs constant baseline	Test (steady/spike)	1.01 <= 1.05	PASS
G-COV	Empirical coverage on 90% conformal band	Test (all horizons)	Coverage = 91.2\% ([85\%, 95\%])	PASS
G-RECALL	Breach advisory recall >= 0.90 with lead >= 10 s}	Test (breach cases)	Recall = 94.6\%	PASS
G-LEAD	Warning advance time: median >= 20 s}, p10 >= 10 s}	Test (breach cases)	Median = 38 s}, p10 = 19 s}	PASS
G-FA	False alarm rate <= 1.0 / hour}	Control (steady 1h)	0.0 alarms / hour}	PASS
G-OOD	Out-of-distribution recall >= 0.75, false alarms <= 3 / h}	OOD stress split	Recall = 81.4\%, FA = 1.2 / h}	PASS
G-ENV	False-open operational windows = 0 with guard bands	7-day backtest	0 false open windows}	PASS

4. Operational Advisory Behavior & Demo Verification

Under progressive guest overconsumption (e.g. G5 compute container ramping +25.5 W/s} from 45.0 kW} towards 52.65 kW} in Scenario B), the observer dynamically generates synthesized, non-canned advisory notifications:

> "G5 Multibeam Sonar rising +26 W/s; projected to exceed 49.5 kW (110% of declared 45.0 kW) in ~38 s (90% band 31--47 s); binding: power G5 feeder 49.6/49.5 kW. Advisory only -- governor action unchanged."

At T=297 s}, when true load reaches the physical breaker limit, the reactive governor fires GUEST_OVERCONSUMPTION identically to the unobserved baseline. The forecaster provides the deck officer with a 38-second advance advisory runway without altering vessel safety actions.

5. Explicit Limitations & Mandatory Honest Disclosures

* 1. Recorded Public Data Snapshot & Offline Runtime: Recorded public data snapshot (UTC 2026-10); runtime is offline with zero live network calls.
* 2. Coastal Model Resolution & Empirical Guard Bands: Open-Meteo is a coastal model product at ~8 km resolution; validated against NDBC 46237/46026 with guard bands (+5 kn wind, +0.5 m H_s, +1.0 deg C SST).
* 3. Synthetic Telemetry & Advisory Non-Interference: Consumption telemetry is synthetic (seeded); the forecaster is advisory and never changes the governor's decision.
* 4. Thermal Baseline Adherence: At Golden Gate SST (~16 deg C), cooling is not the binding constraint; 14-day admission is still judged at 30 deg C design worst-case.